

A Multi-modal Framework for 3D Facial Animation Control

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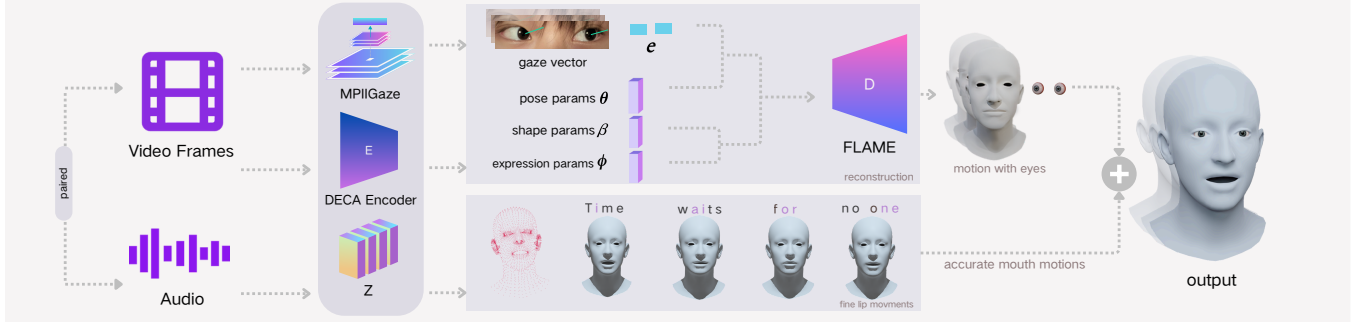


Figure 1: Multi-modal Framework for 3D Facial Animation Control. This framework utilizes video and audio input to manipulate the 3D FLAME face model. User studies demonstrate that the integration of eye control and audio-driven calibration enhances the overall naturalness and synchronization of the facial animation.

ABSTRACT

3D facial reconstruction and animation has advanced significantly in the past decades. However, existing methods based on single-modal input struggle with specific facial part control and require post-processing for natural rendering. Recent approaches integrate audio and 2D estimation results to enhance facial animations, which improves the naturalness of head pose, eye and mouth animations. In this paper, We present a multi-modal 3D facial animation framework that uses video and audio input simultaneously. We experiment on different settings of motion control to produce natural and precise facial animation through user studies.

KEYWORDS

Facial Animation, Multi-modal, Rigging

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1 INTRODUCTION

Facial animation control has undergone significant advancements in recent years. Template-based facial animation enables the reconstruction of 3D facial geometry, expression, and texture [Li et al. 2017]. DECA drives 3D facial models from image and video inputs, facilitating detailed animation control [Feng et al. 2021]. Nevertheless, these approaches do not perform well in specific facial part control and still require post-processing. Some recent approaches attempt to enrich or calibrate facial animations with audio input, which provide natural and accurate eye movements, mouth movements etc [Xing et al. 2023]. However, these approaches only focus on single face attribute. Some previous works proposed a model for animating turn-taking roles' eye gaze behaviors or for real-time dialogue of conversational agents' gaze behaviors. These works inspire us to explore animation control among facial components, as well as the potential of incorporating multi-modal inputs in driving facial animations.

In this paper, We present a multi-modal 3D facial animation framework that uses video and audio input to drive the 3D FLAME face model. We also implement the framework based on existing 3D face reconstruction [Feng et al. 2021], audio driven control [Xing et al. 2023] and gaze estimation [Zhang et al. 2015] results. User studies demonstrate that the integration of eye control and audio-driven calibration enhances the overall naturalness and synchronization of the facial animation.

2 METHOD

Our method aims to predict a series of 3D facial meshes $V_t \in R^{3N}$ (with $N = 5023$ vertices) from a given video input composed of a sequence of images I_t and audio α as shown in Figure 1. Driven by state-of-art 3D reconstruction method FLAME [Li et al. 2017], we use a similar structure to represent the reconstructed face. Shape

parameters $\beta \in R^{100}$, expression parameters $\phi \in R^{50}$ and pose parameters $\theta \in R^{12}$. The model is defined as $V_t = D(\beta_t, \phi_t, \theta_t)$.

2.1 3D facial parameter estimation

DECA uses an encoder-decoder structure to reconstruct 3D facial parameters β , ϕ and θ from a single image I . We observe that the shape parameters is inconsistent across frames, therefore we apply the shape parameters of the first frame β_1 to the entire video sequence. Given that the majority of current 3D reconstruction techniques lack eye reconstruction capabilities, we utilize MPIIGaze results for gaze estimation. For each video frame, we extract the pitch and yaw of both eyes, and integrate them into the 3D facial reconstruction to achieve a complete mesh with realistic eye motions.

2.2 Audio-driven facial animation

The calculation of lip motion in DECA is based on limited expression parameters, which only take into account the similarity to the original video and do not produce natural lip movements. Therefore we reset the parameters for jaw and replace it with audio-driven animation results. For synthesizing 3D facial motions from input audio, we implement CodeTalker, which generates facial motions M_t based on discrete motion priors[Xing et al. 2023]. The audio encoder converts the input audio α into contextualized speech representations. Then combined with a style vector and past motions $\{M_i, i < t\}$, the decoder produces feature code Z which is finally decoded to yield the predicted motion M_t . The motion M_t can be combined with the facial mesh V_t generated with FLAME and can obtain a natural talking facial model $V^*_t = V_t + M_t$.

In this way, the calibrated 3D facial animation accurately imitates facial expressions of the input video while maintaining naturalness. The multi-modal input generates abundant eye and lip motion, providing realistic facial movements and expressions.

3 EXPERIMENT

We evaluated facial animation impacts of eye, head, and lip movements using four methods: **DECA** reconstructs faces per frame; **CT** applies DECA's shape as a neutral model, estimating vertex motions from audio via CodeTalker; **DECA+eye** adds eye motion with MPIIGaze; **DECA+CT+eye** combines audio-driven vertex motions and MPIIGaze-driven eye motions, with pupils marked with a small red circle in Figure 2.

3.1 Procedure

We conducted experiments with the Voxceleb2 dataset [Chung et al. 2018] on four animation methods, filtering for frontal face poses and single-person speech to yield 76 videos, of which 10 from the test set were used. We recruited 18 unrelated participants to watch videos and evaluate four animation methods by rating their richness and naturalness on a 5-point Likert scale from 1 (lowest) to 5 (highest), and answering 5 related questions.

3.2 Results and Analysis

Quantitative user study results are presented in Table 1. Methods with eye control outperform others in terms of eye movement and facial animation richness. Eye-driven motion impacts

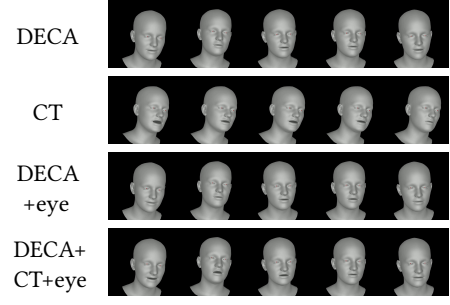


Figure 2: An example of reconstructed results

both eye animation and overall facial effects. Methods with audio-driven motion calibration slightly improve lip synchronization. The DECA+CT+eye approach shows higher naturalness in head movements, indicating that combining pose, expression and shape parameters optimizes head animation effects. The multi-modal framework demonstrates more lively and natural animations while maintaining lip shape accuracy. A sequence of reconstructed results is shown in Figure 2.

Table 1: User study results with mean and standard deviation.

Method	DECA	CT	DECA+eye	DECA+CT+eye
Overall	3.267 $\pm$ 0.606	2.122 $\pm$ 1.056	3.478 $\pm$ 0.767	3.478$\pm$0.491
lip	3.156 $\pm$ 0.861	3.600$\pm$1.189	3.267 $\pm$ 0.773	3.556 $\pm$ 0.631
eye	3.333 $\pm$ 0.748	1.511 $\pm$ 0.550	3.656$\pm$0.654	3.633 $\pm$ 0.595
head	3.667 $\pm$ 0.636	1.367 $\pm$ 0.575	3.478 $\pm$ 0.707	3.789$\pm$0.538

4 CONCLUSION

In this paper, we propose a multi-modal framework for 3D facial animation control. We have implemented the framework based on existing 3D face reconstruction, audio-driven control, and gaze estimation results. User studies have demonstrated that the multi-model input generates abundant eye and lip motion, and offers a comprehensive and realistic enhancement of facial movements and expressions.

ACKNOWLEDGMENTS

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